YAKEEN

NEET 2.0 2027

Physics | Lecture - 5

Basic Maths and Calculus

By- Manish Raj (MR Sir)



Today's Goal

PhD on straight line

Hello sir ji me yuganda se bol raha hu ..me yakeen 2.0 se hu ...mera ak suggestions tha ...aap ak kaam kar sakte he ki - jese aap har week jitni bhi class le, topics, question jo bhi kahai, wo week ke end me means Saturday evening ko uus week ka ak practice paper de sakte he jisme uus week jo jo parhaya tha wo sab topics ke question rahega ..jisse hamari bohot help hogi and revision me ham last me wo questions practice karke ja sakte he

Agar atcha lage to kal class me boliyega and hamko chemistry sir ko bhi bola tha unhone bhi karenge bola

$y = 8 - 6\cos x$

$y_{max} = 8 - 6(\cos x)_{min}$

$= 8 - 6(-1)$

$= 8 + 6$

$= 14$

$y_{min} = 8 - 6(\cos x)_{max}$

$= 8 - 6(1)$

$= 2$

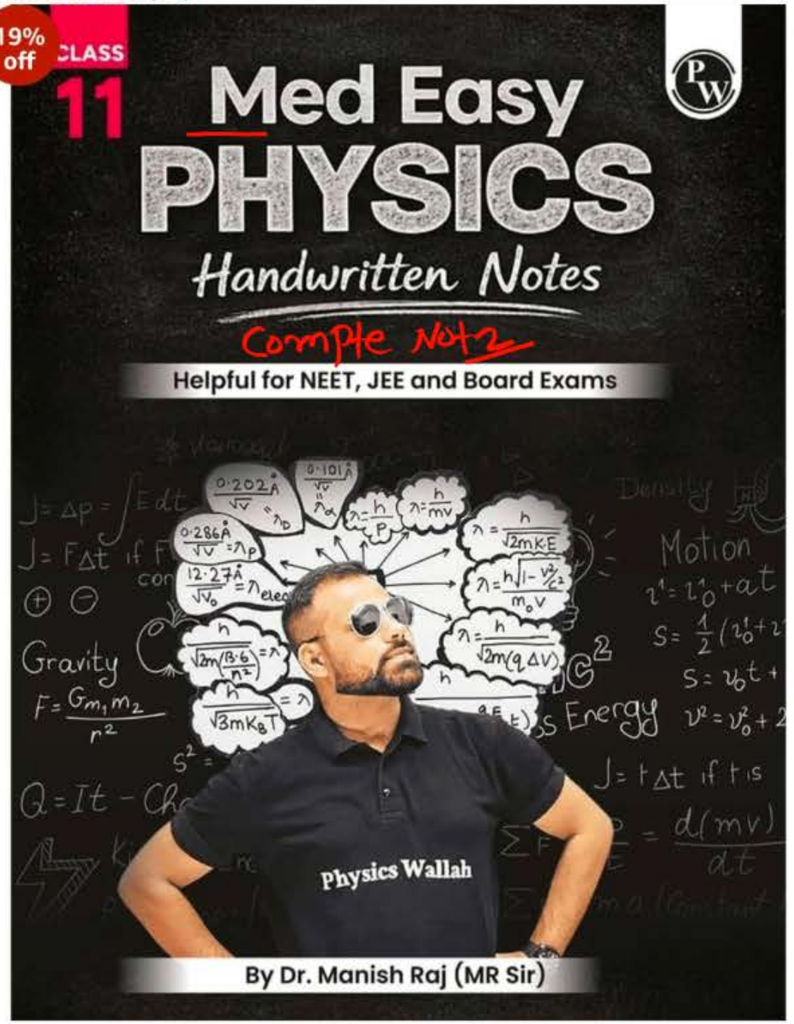
10:29 PM

Good evening sir
Sir mai yakeen 2.0 2027 ka student hu sir plz class mai ya Q explain kar dejiyaga
Q)
Find maximum and minimum value of 8-6cosx
SIR mera max 2 aa raha hai aur min 14

Double-tap to ❤️

Message...

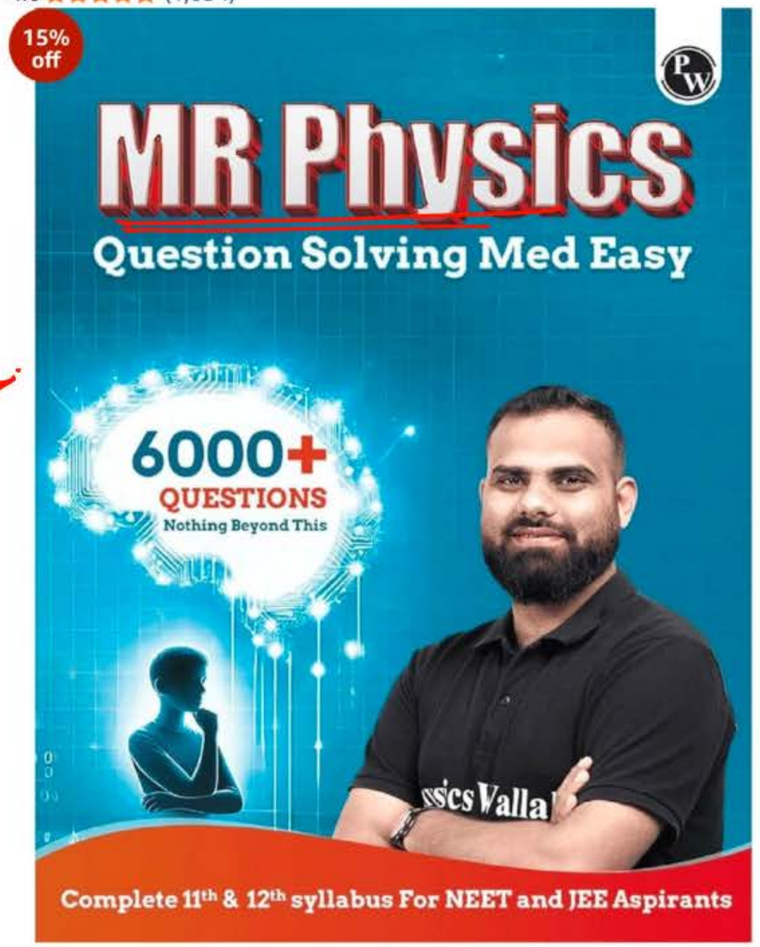
4.6 ★★★★½ (13)



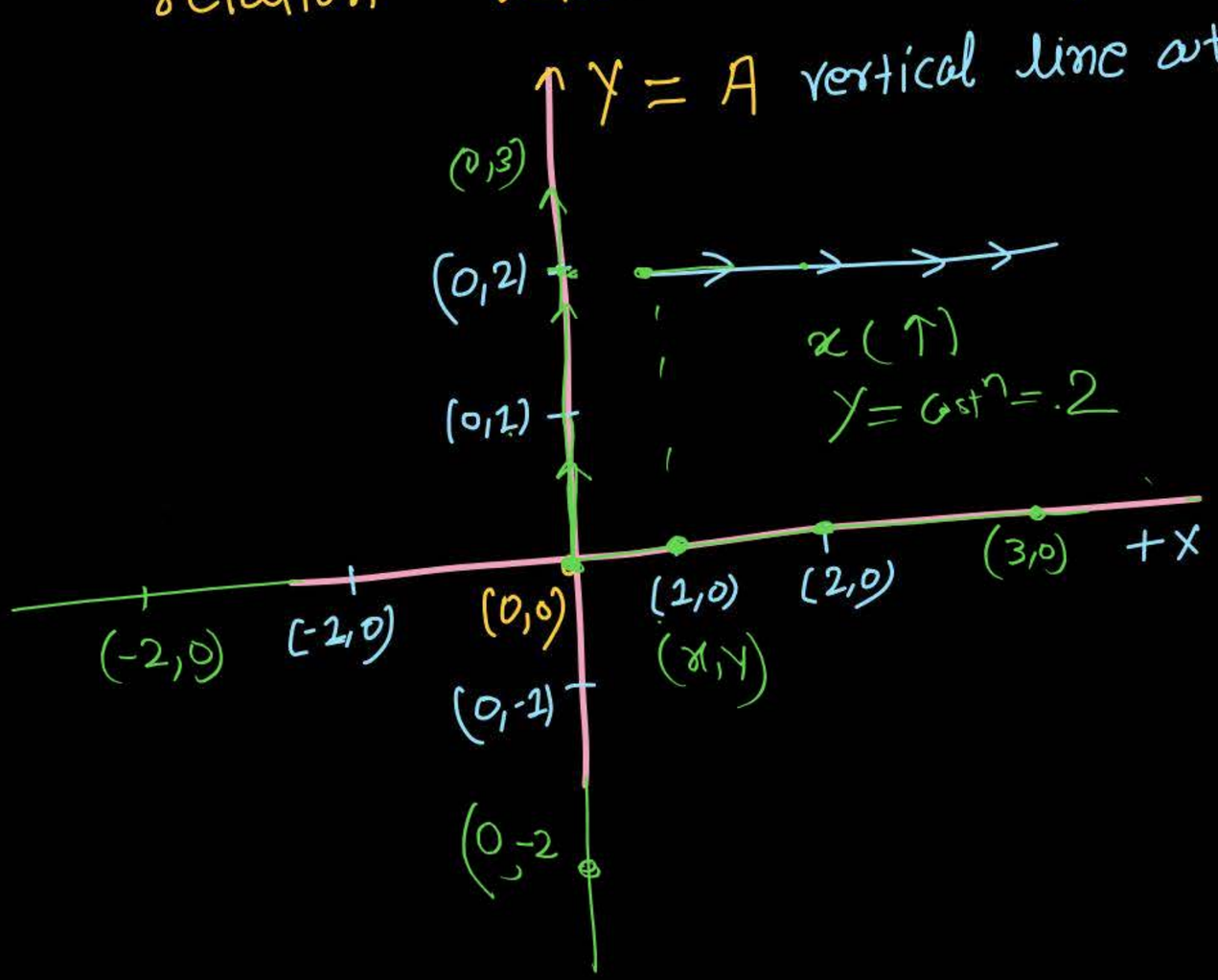
5 July 2025

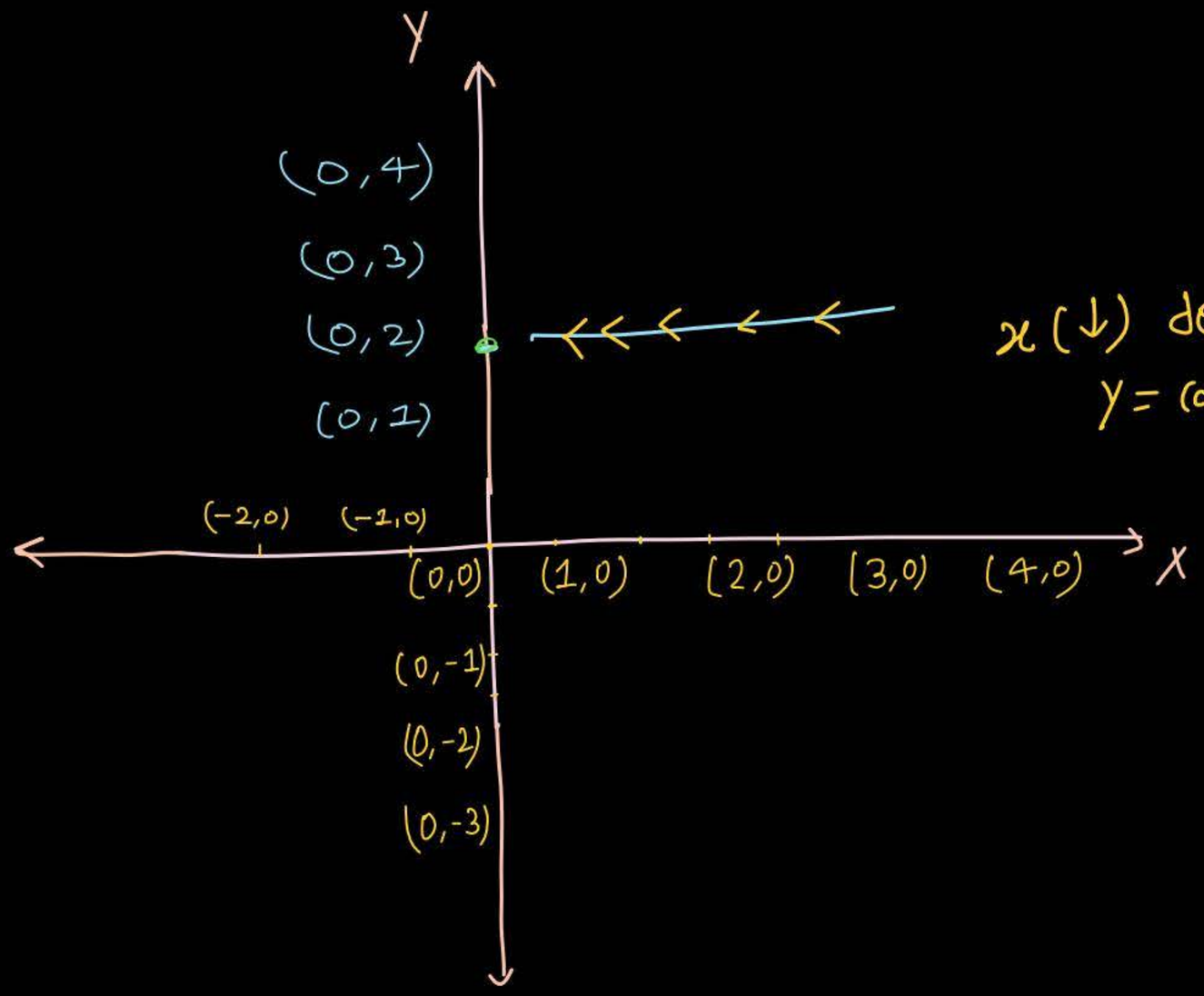
ISBN-13: 978-9371531382 **ISBN-10:** 937153138X

4.6 ★★★★½ (1,054)



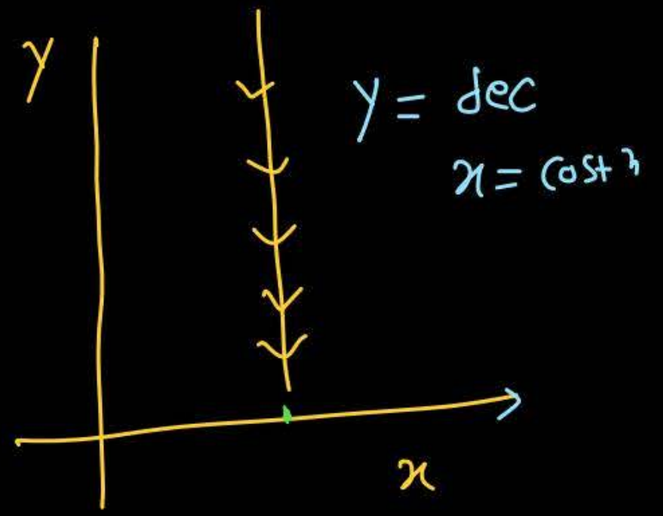
Graph ⟶ A diagram in which a line or curve shows the relation between two physical quantity.

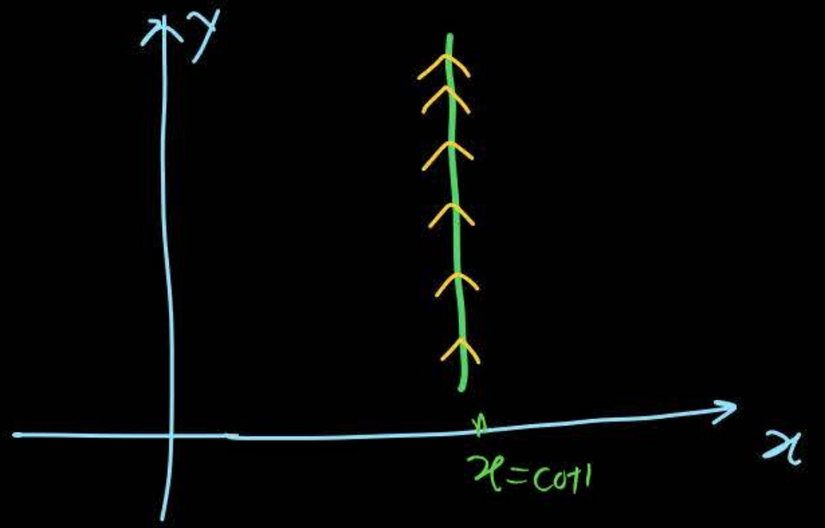




x (↓) dec

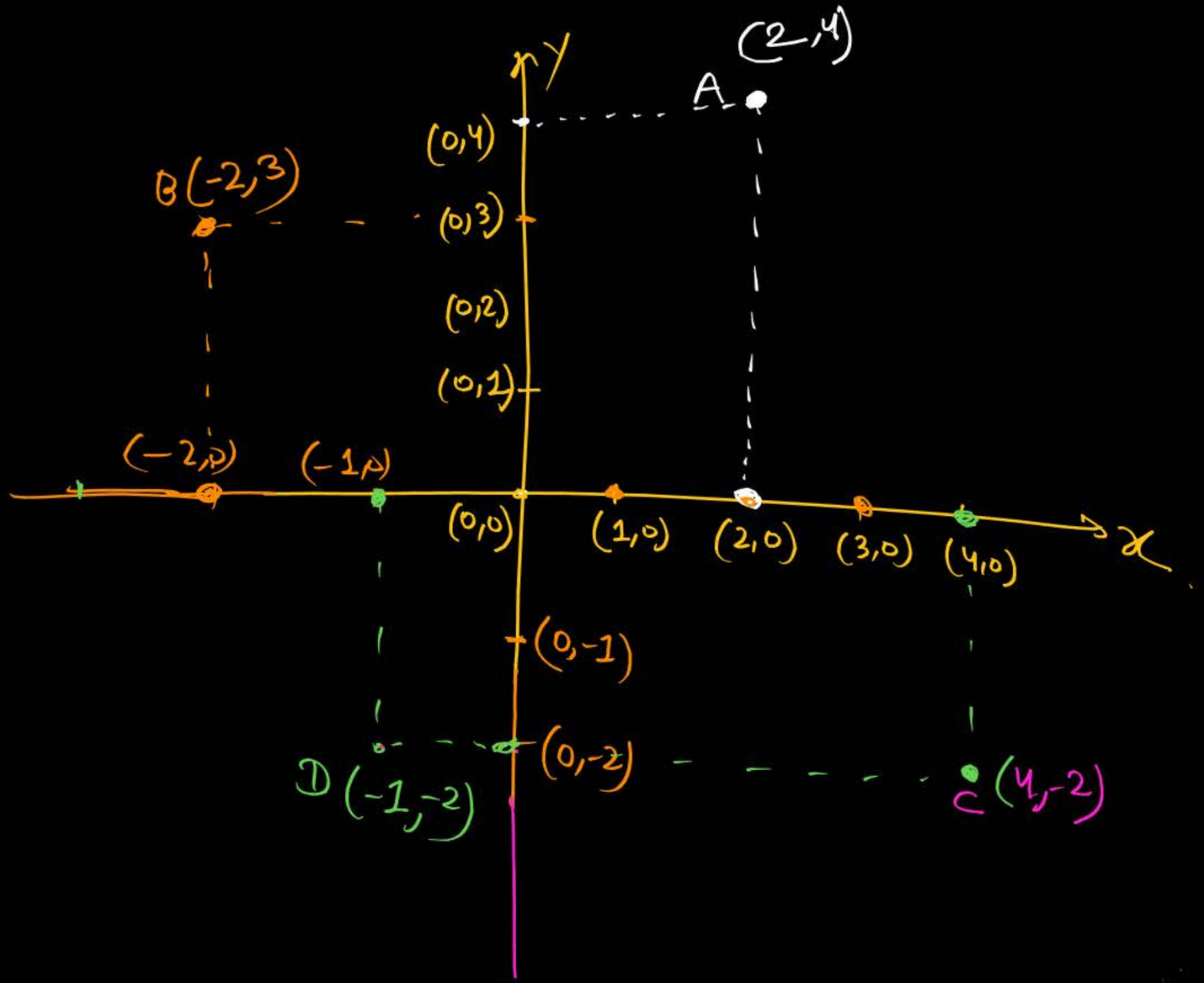
$y = \text{cost}^n = 2$



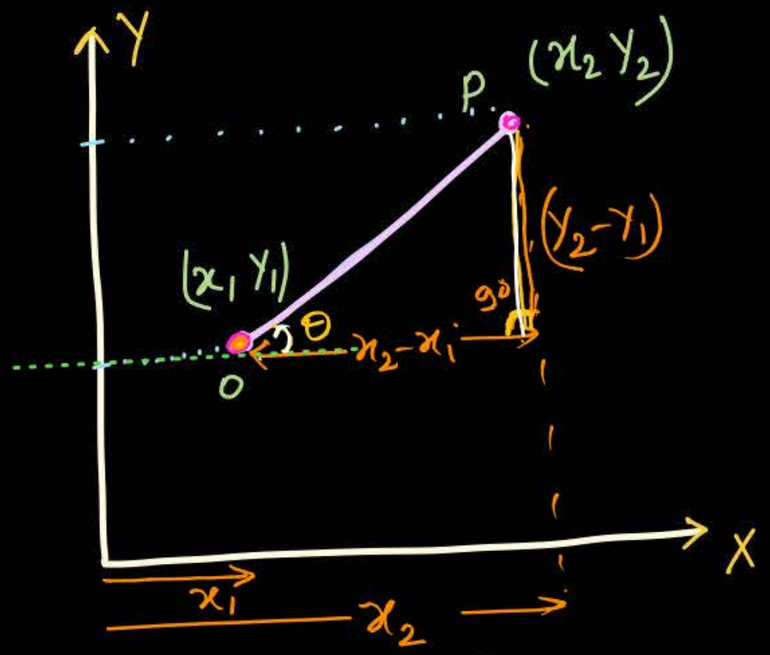


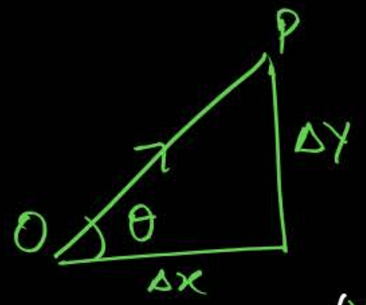
Copy

Locate Point A(+2,+4) B(-2,3) C(+4,-2) D(-1,-2)



Distance formula





$$m = \tan\theta = \frac{\Delta y}{\Delta x} = \frac{(y_2-y_1)}{(x_2-x_1)} = \frac{\text{change in } y}{\text{change in } x}$$

slope = $\tan\theta$ = change in y w.r.t change in x.

$$OP^2 = (x_2-x_1)^2 + (y_2-y_1)^2$$

$$OP = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2}$$

↳ of all three co-ordinate are given

- $O(x_1, y_1, z_1)$
- $P(x_2, y_2, z_2)$

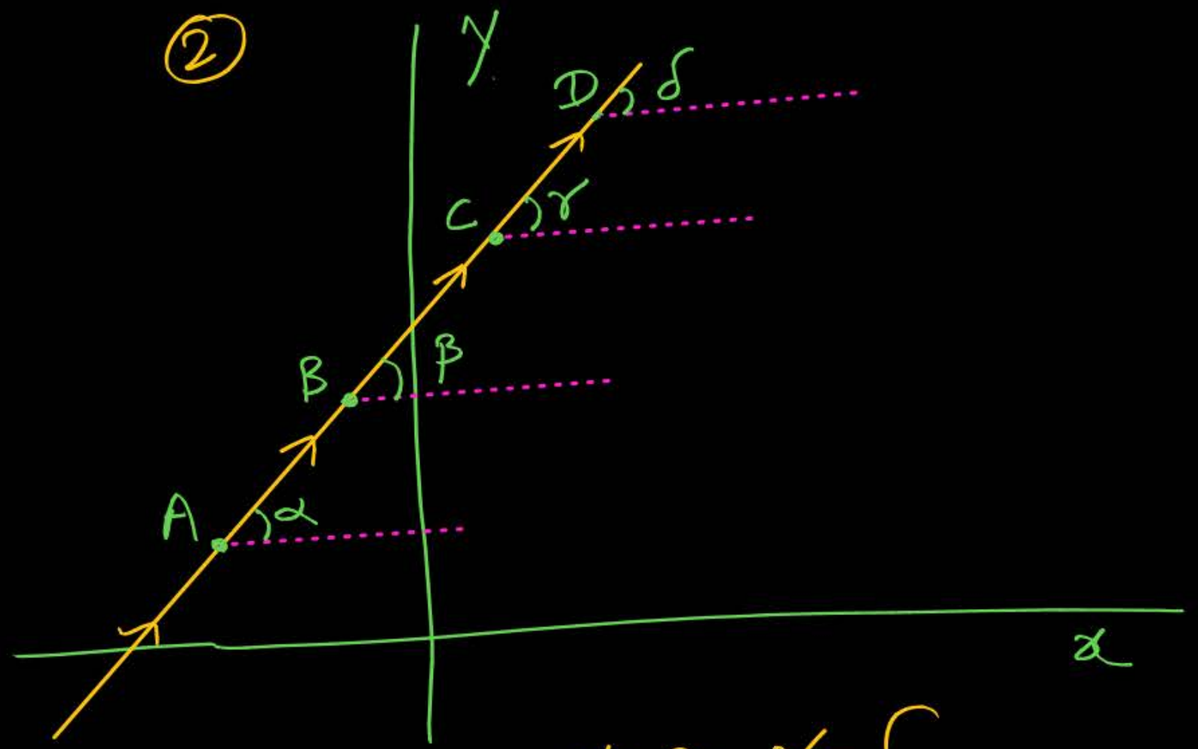
$$OP = \sqrt{(x_2-x_1)^2 + (y_2-y_1)^2 + (z_2-z_1)^2}$$

① If $P(2,-1,0)$ & $Q(3,1,2)$ find distⁿ b/w P & Q

$$PQ = \sqrt{(3-2)^2 + (1-(-1))^2 + (2-0)^2}$$

$$= \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3$$

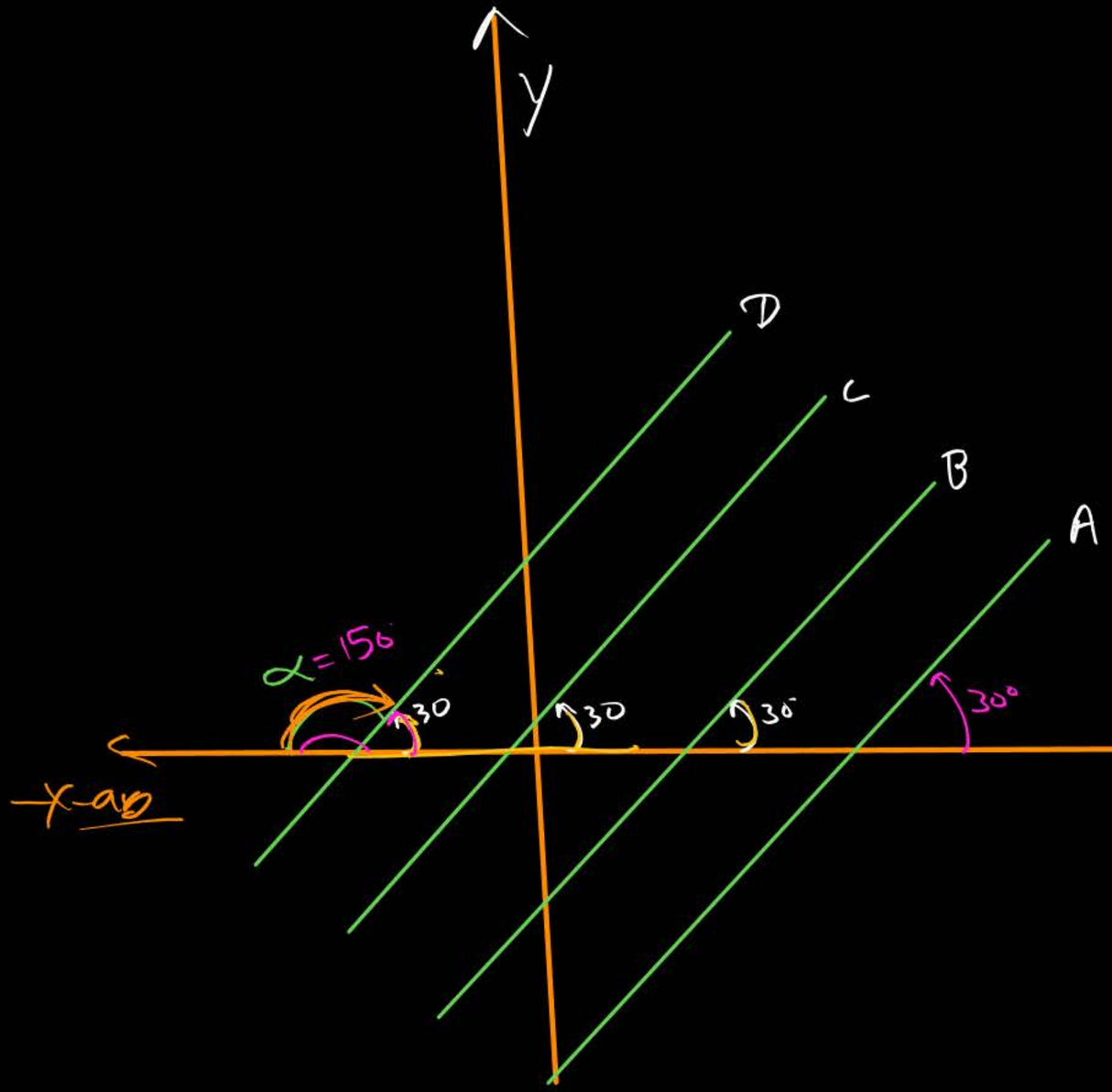
(2)



$\alpha = \beta = \gamma = \delta$

$\tan\theta = \text{Sam}$

Slope same at all point.



(3) Compair Slope of these Parallel line

$m_A = m_B = m_C = m_D = \tan 30^\circ$

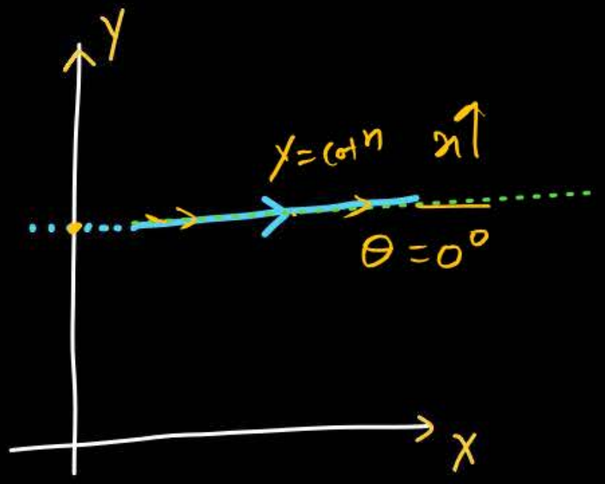
$= \frac{+1}{\sqrt{3}}$ ✓

Slope $= \tan\theta$ ← Angle from +ve X-axis in Anti clock

Slope $= -\tan\alpha$ ← Angle from -ve X-axis

Slope $= -\tan(150^\circ) = -\left(\frac{-1}{\sqrt{3}}\right) = \frac{+1}{\sqrt{3}}$

④



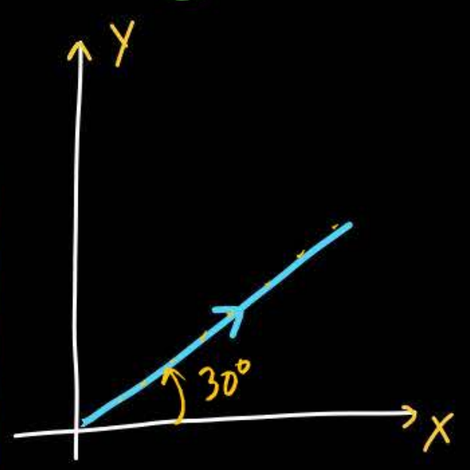
$$\text{Slope} = \tan 0^\circ = 0$$

- A line ∥ to X-axis have zero slope

$$\text{Slope} = \tan\theta = \frac{\Delta Y}{\Delta x} = \frac{0}{\Delta x} = 0$$

$y = \text{constant}$, $\Delta y = 0$

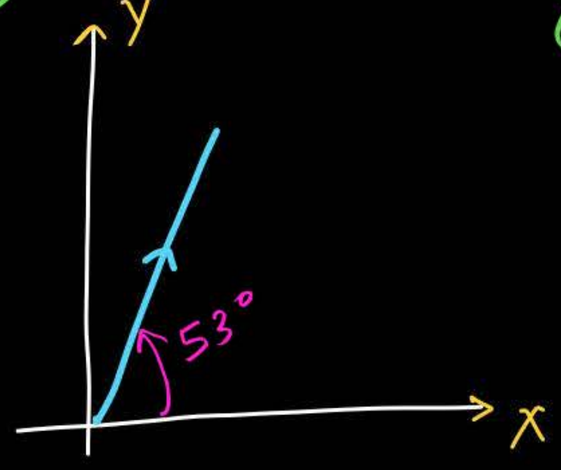
⑤



$$m = \tan\theta = \tan 30^\circ$$

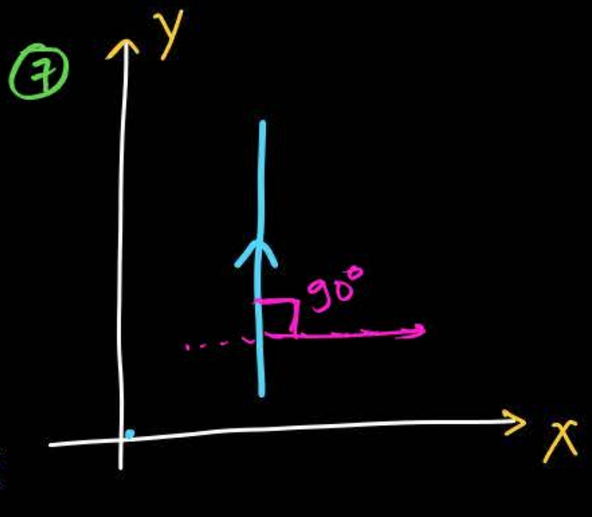
$$\text{Slope} = \frac{1}{\sqrt{3}}$$

⑥



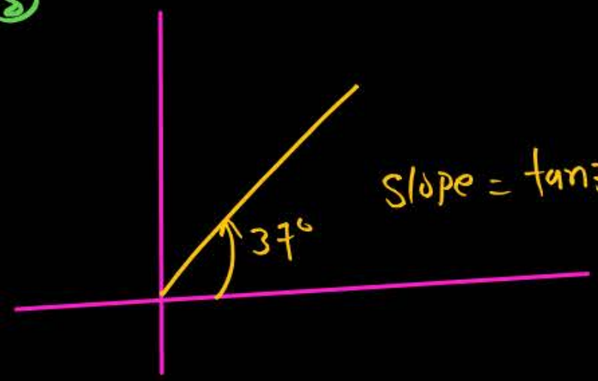
$$m = \tan 53^\circ = \frac{4}{3}$$

⑦



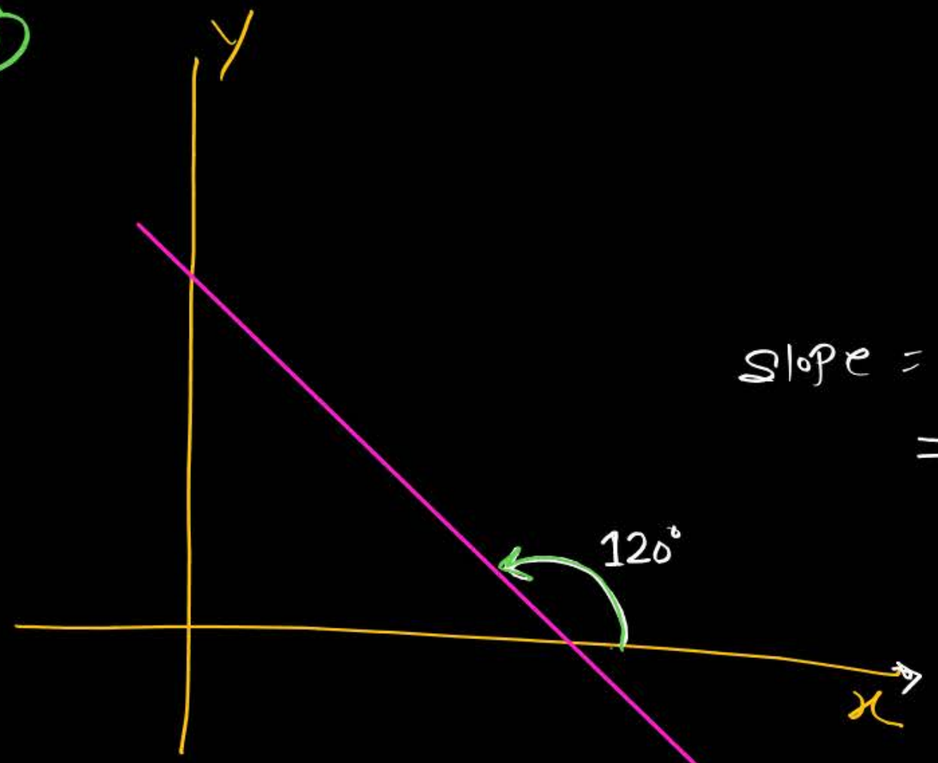
$$\text{Slope} = m = \tan 90^\circ = \frac{1}{0} = \text{infinite}$$

⑧

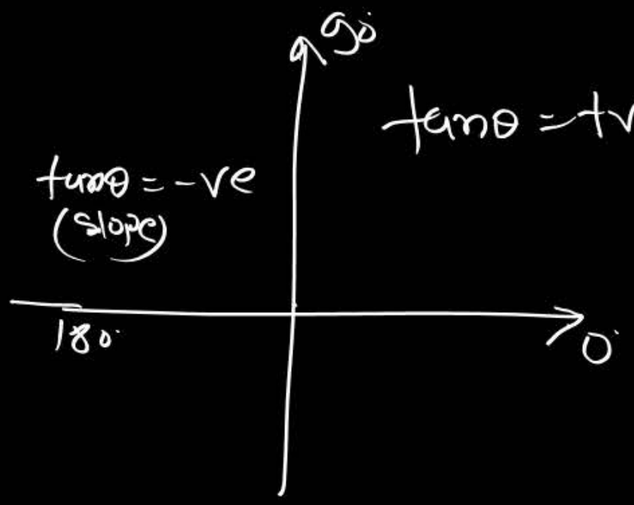


$$\text{Slope} = \tan 37^\circ = \frac{3}{4}$$

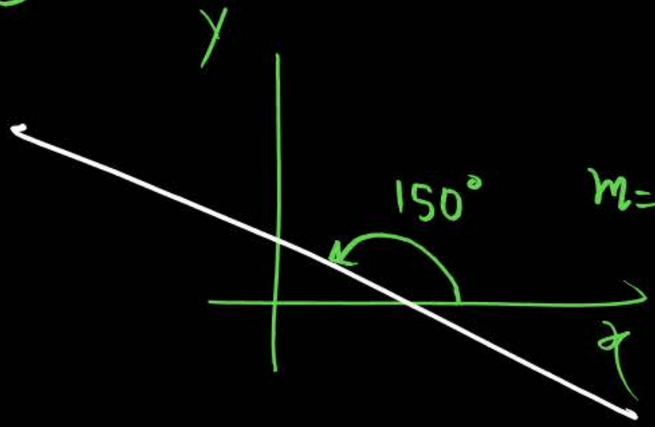
(9)



$$\text{slope} = \tan(120^\circ)$$
$$= -\sqrt{3}$$



(10)



$$m = \tan(150^\circ) = -\frac{1}{\sqrt{3}}$$

MR* Box

- If Angle made by +ve x-axis is acute then slope +ve

$\{0° \leq \theta < 90°\}$

If Angle made by +ve axis is obtuse; m = -ve

$[90° < \theta \leq 180°]$

If $\theta = 0°$

$m = 0$

$\theta = 90°$

m = infinite

MR Scam

(11) Angle made by straight line with +x axis is called slope. (in Anti-

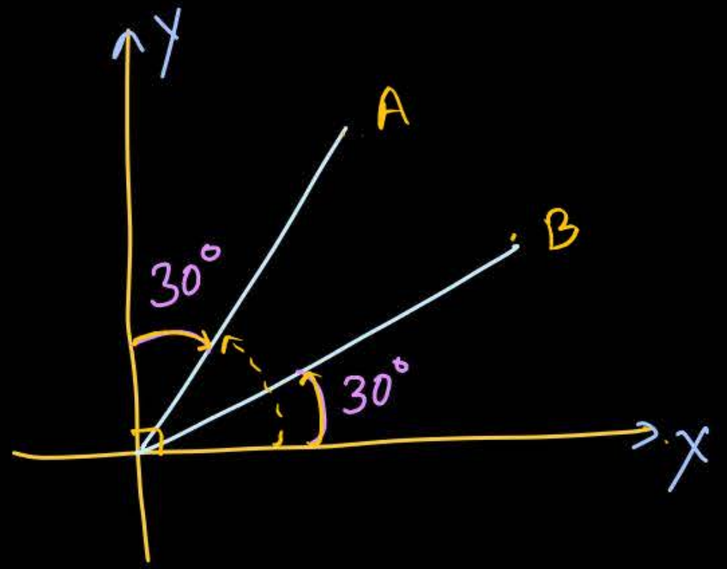
(a) True (52%)

(b) false (46%)

Slop = θ ✗

slope = $\tan\theta$ ✓

(12)

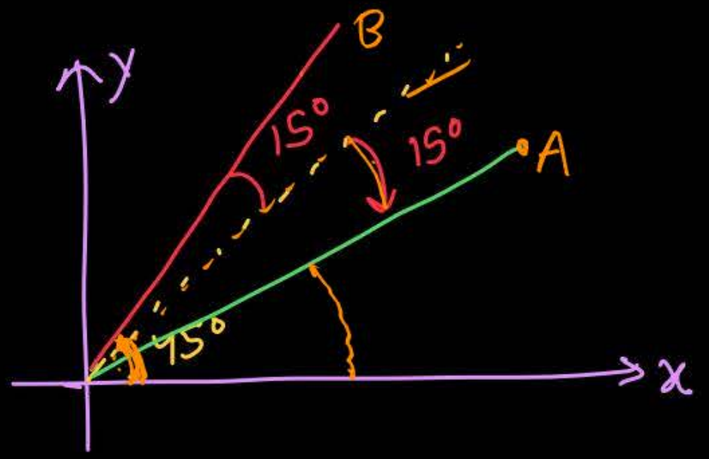


find ratio of slope.

$$\frac{(\text{Slope})_A = \tan 60^\circ}{(\text{Slope})_B = \tan 30^\circ}$$

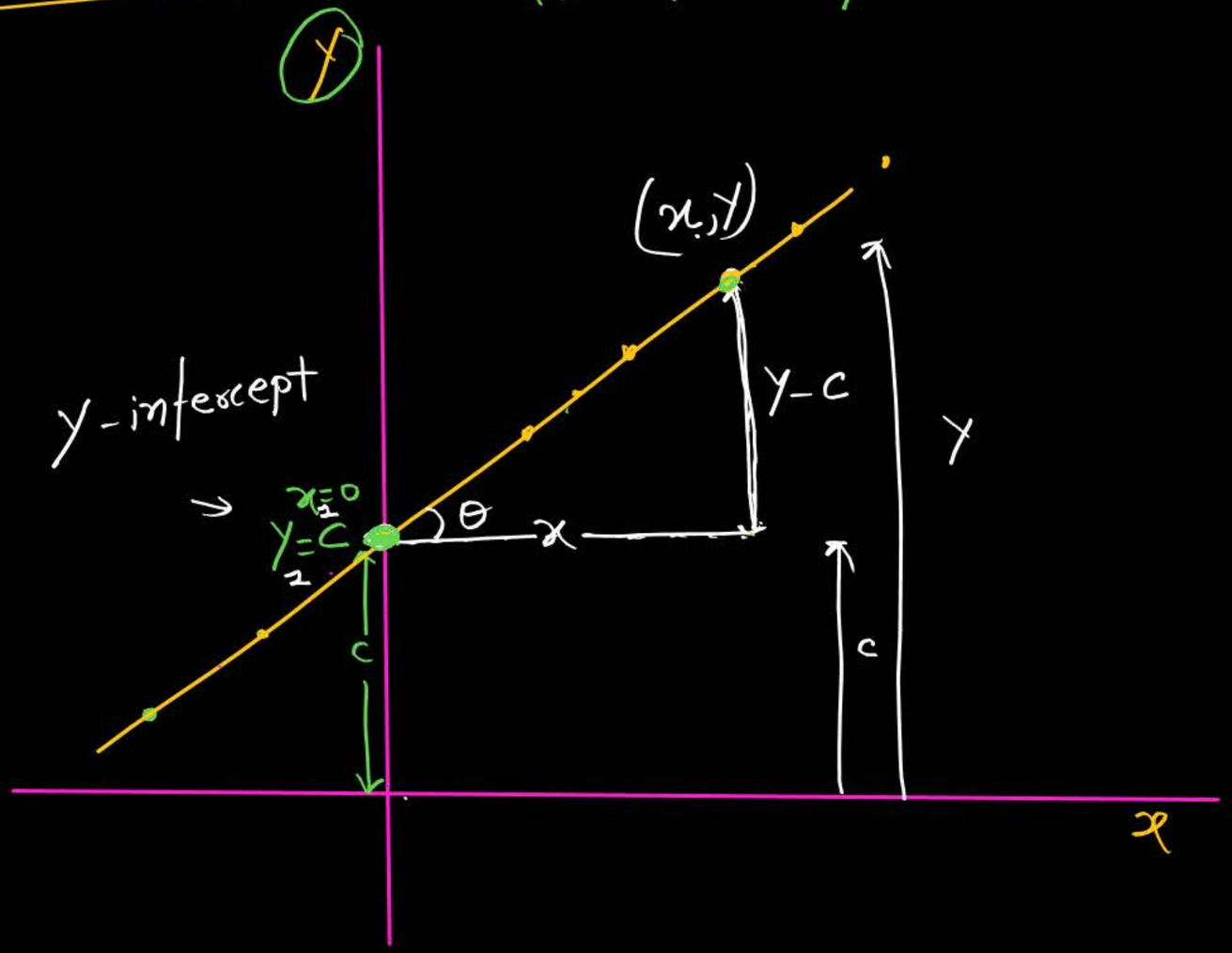
$$\frac{m_A}{m_B} = \frac{\sqrt{3}}{1/\sqrt{3}} = \frac{\sqrt{3}\times\sqrt{3}}{1} = \frac{3}{1}$$

(13)



$$\frac{m_A}{m_B} = \frac{\tan 30^\circ}{\tan 60^\circ} = \frac{1}{\sqrt{3}\times\sqrt{3}} = \frac{1}{3}$$

Eqn of straight line = Relation b/w x & y



$m = \tan\theta$

$m = \frac{y-c}{x}$

$y - c = mx$

$y = mx + c$

Rel^n B/w 'x' & 'y'

m = slope

y-Intercept = c (at x=0)

COPY

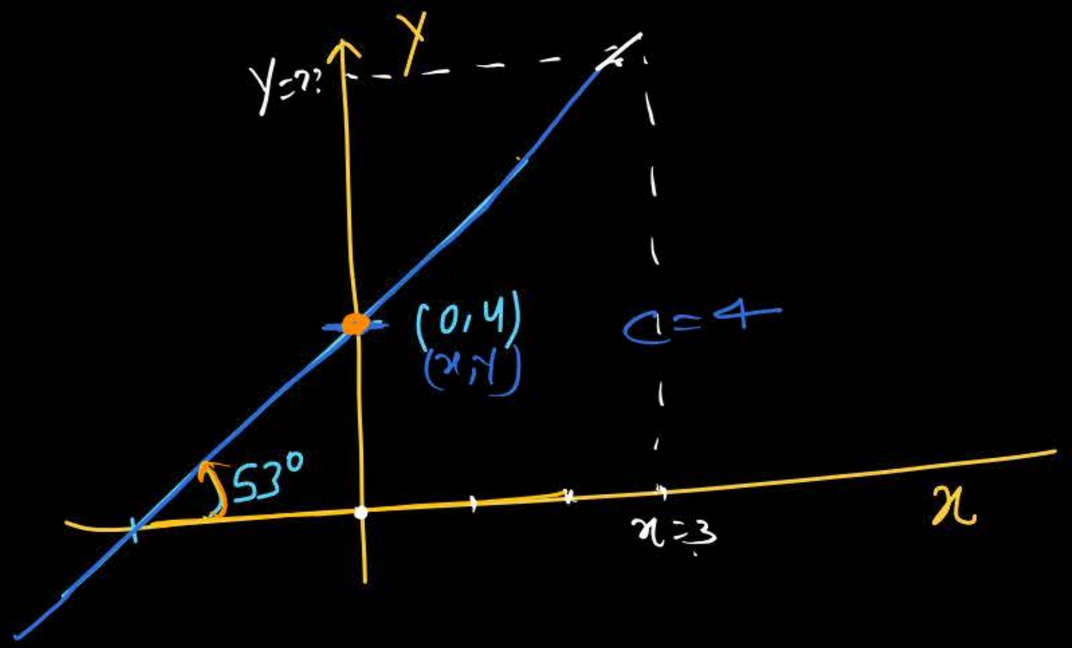
MR* Box

- Straight line ka ek hi nara 'c' and 'm' fixed ho hamara.

- Tum mujhe merc 'c', 'm' do mai (st. line) kisi bhi point pe x and y ka relation dunga.

$$y = mx + c$$

Write equation of straight line & find value of y' at $x=3$



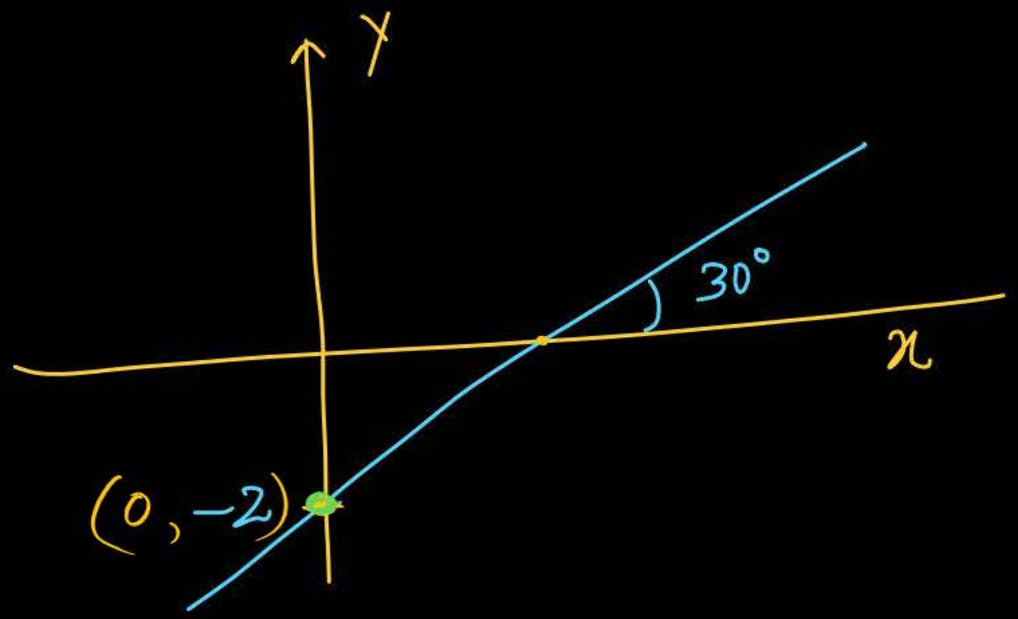
Slope $= \tan 53^\circ = \frac{4}{3}$

$y = mx + c$

$$y = \frac{4}{3}x + 4$$

$$y = \frac{4}{3} \times 3 + 4 = 8$$

Write equation of straight line, find y at $\boxed{x = 2\sqrt{3}}$



$c = -2$

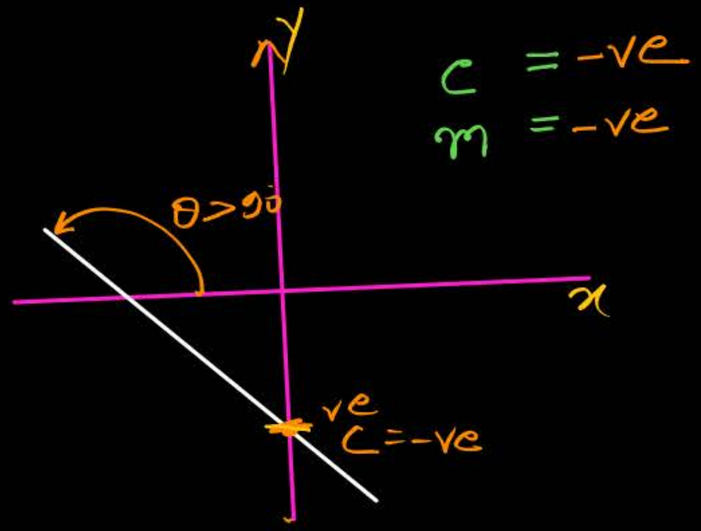
$m = \tan 30^\circ = \frac{1}{\sqrt{3}}$

$y = mx + c$

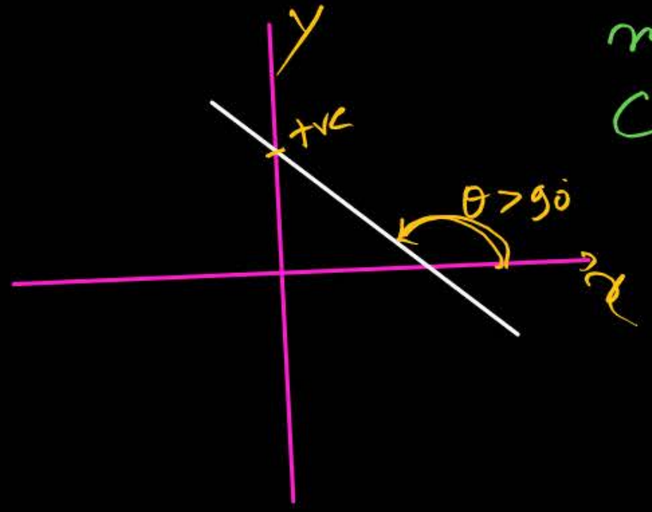
$y = \frac{1}{\sqrt{3}} 2\sqrt{3} - 2$

$y = 2 - 2 = 0$

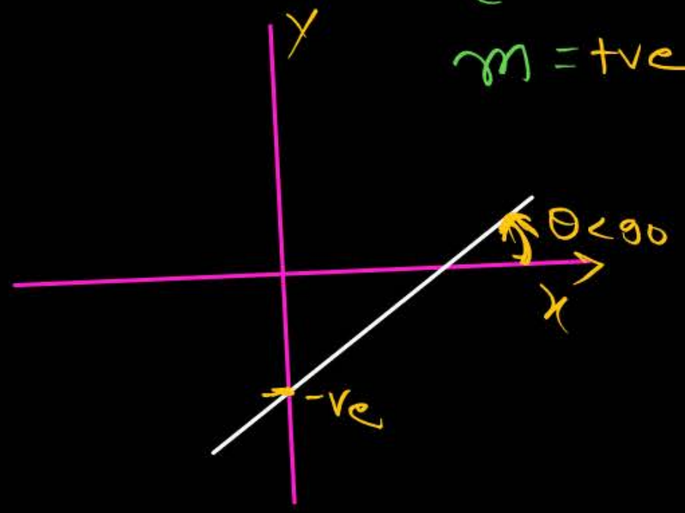
0



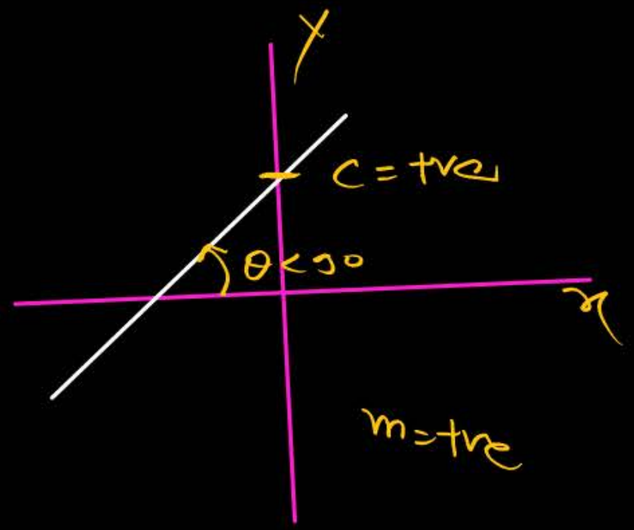
C = -ve
m = -ve



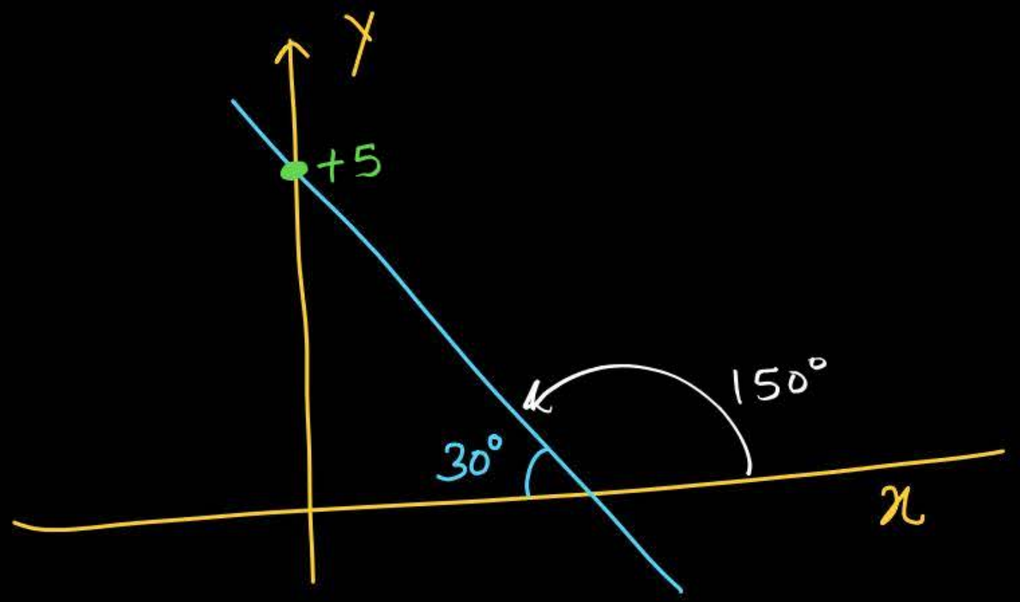
m = -ve
C = +ve



C = -ve
m = +ve



Write equation of straight line, find y at $x = -4$



$$m = \tan(150^\circ) = -\frac{1}{\sqrt{3}}$$

$$C = 5$$

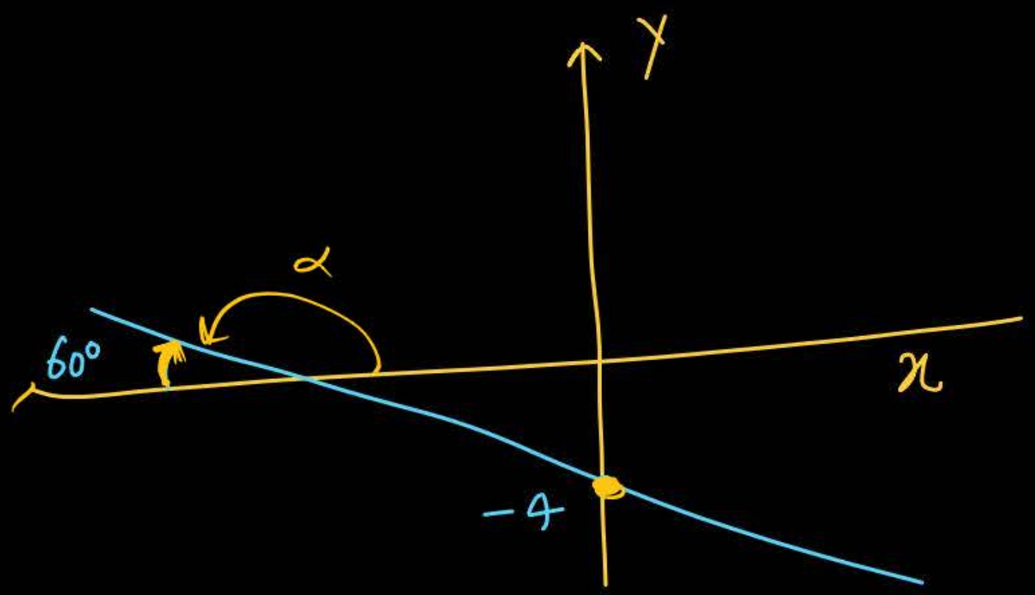
$$y = mx + C$$

$$\boxed{y = -\frac{1}{\sqrt{3}}x + 5}$$

$$y = +\frac{4}{\sqrt{3}} + 5$$

at $x = -4$

write equation of straight line



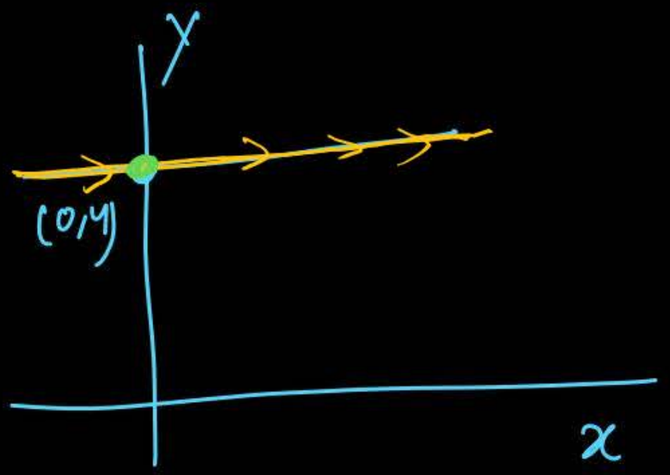
$$C = -4$$

$$m = \tan(120^\circ) = -\sqrt{3}$$

$$y = mx + c$$

$$\boxed{y = -\sqrt{3}x - 4}$$

eqn of straight line



$m = 0$

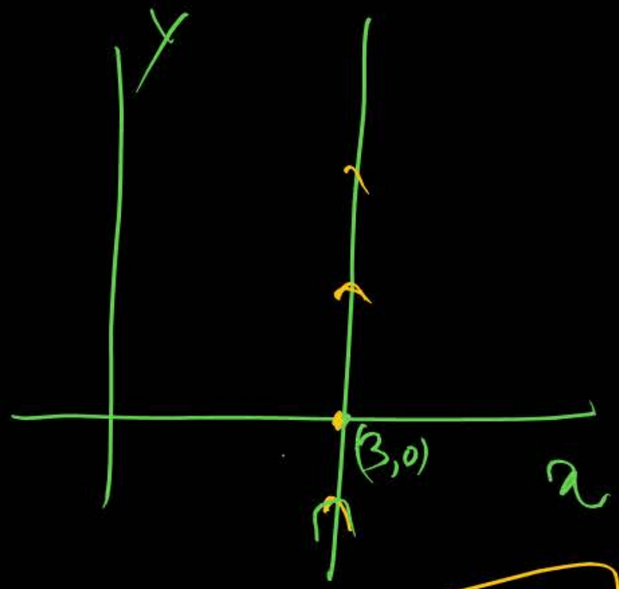
$c = 4$

$y = mx + c$

$y = 0 + 4$

$\boxed{y = 4}$

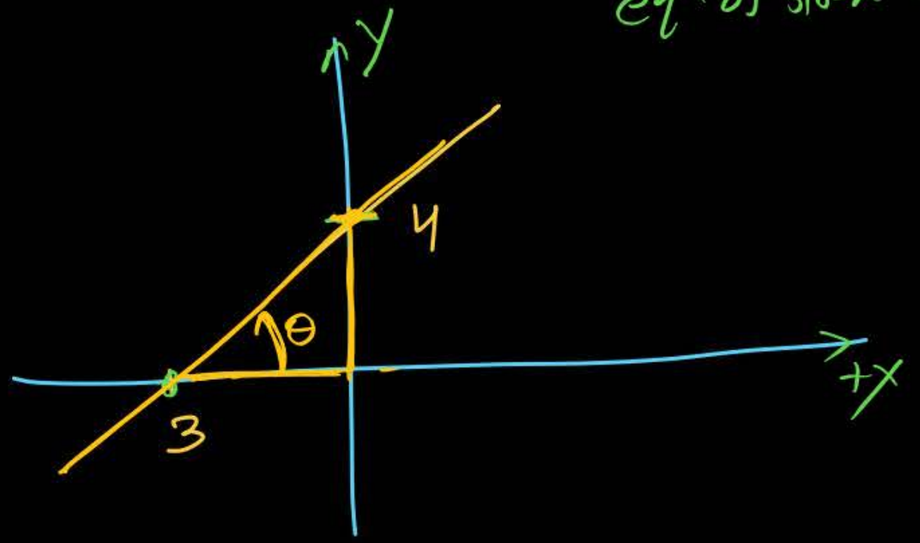
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$\boxed{x = 3}$

#

eqn of straight line.



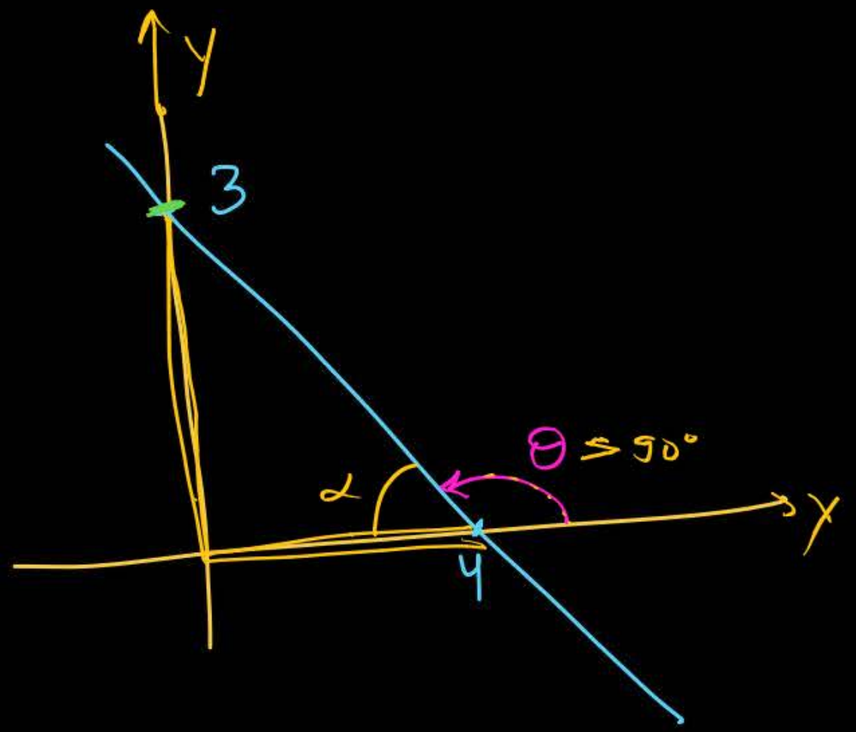
$m = \tan\theta = \frac{\Delta y}{\Delta x}$ ✓

$c = 4$ ✓

$m = \tan\theta = \frac{4}{3}$

$y = mx + c$

$y = \frac{4}{3}x + 4$ ✓



$c = 3$ ✓

$m = -\tan\alpha = -\frac{3}{4}$ ✓

$$y = -\frac{3}{4}x + 3$$

✓ Draw graph of given straight line

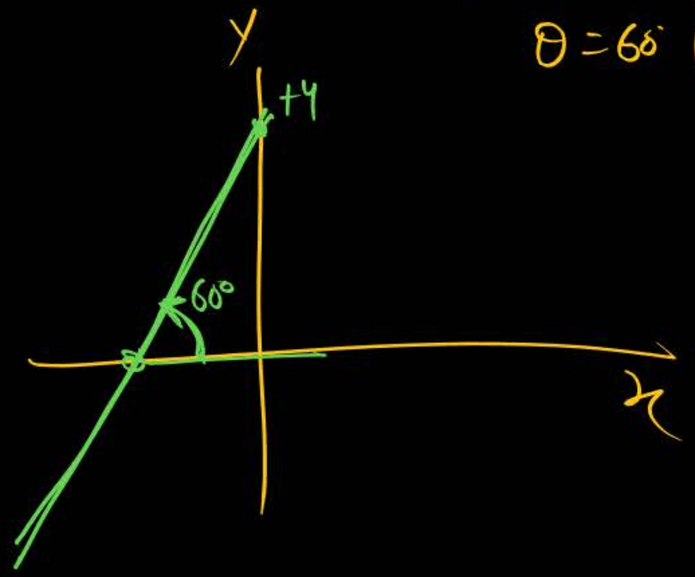
$Y = \sqrt{3}x + 4$

$Y = mx + c$

$c = 4$

$m = \sqrt{3} = \tan\theta$

$\theta = 60°$ (with +x axis)



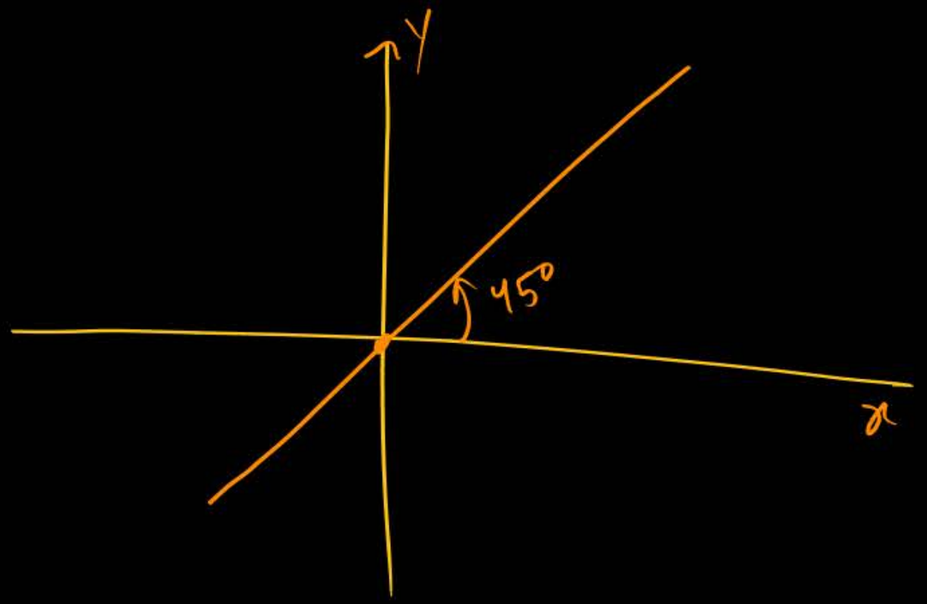
$\Rightarrow$ "$Y = x$"

$Y = mx + c$

$c = 0$

$m = 1 = \tan\theta$

$\theta = 45°$



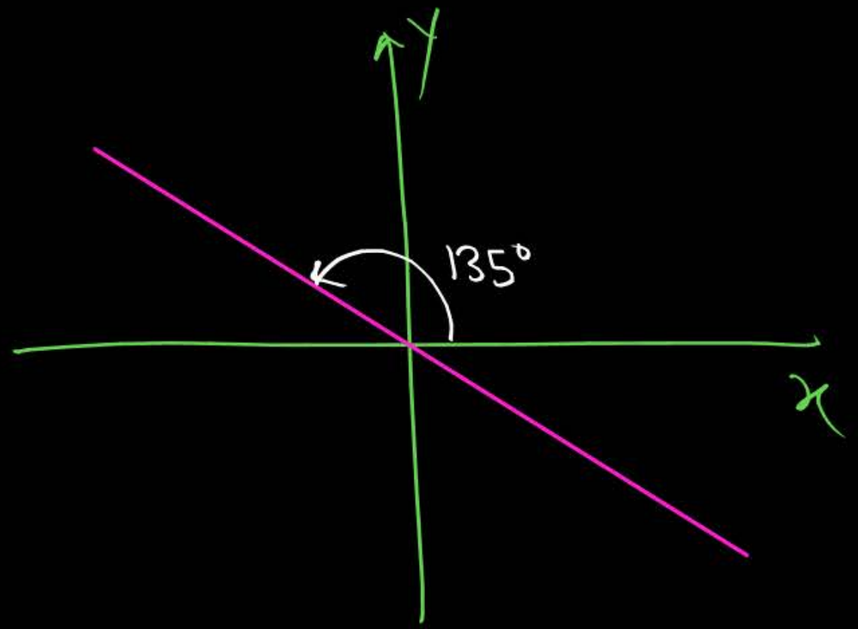
"$y = -x$" ↺ compare

$y = mx + c$

$c = 0$

$m = -1 = \tan\theta$

$\theta = 135°$



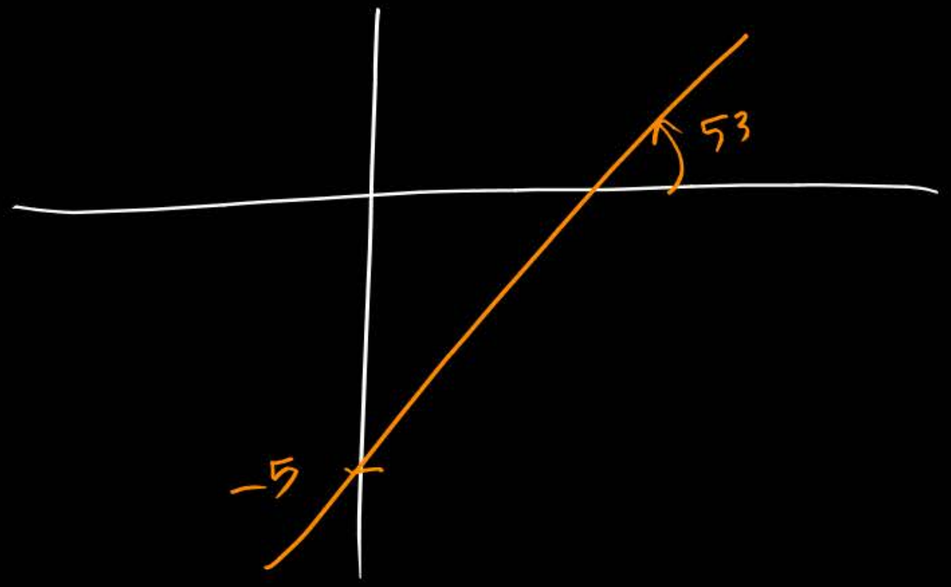
$y = \frac{4}{3}x - 5$ ↻ Comp.

$y = mx + c$

$c = -5$

$m = \frac{4}{3} = \tan\theta$

$\theta = 53°$



"$y = 2x - 1$"

$y = mx + c$

$c = -1$ → slope +ve

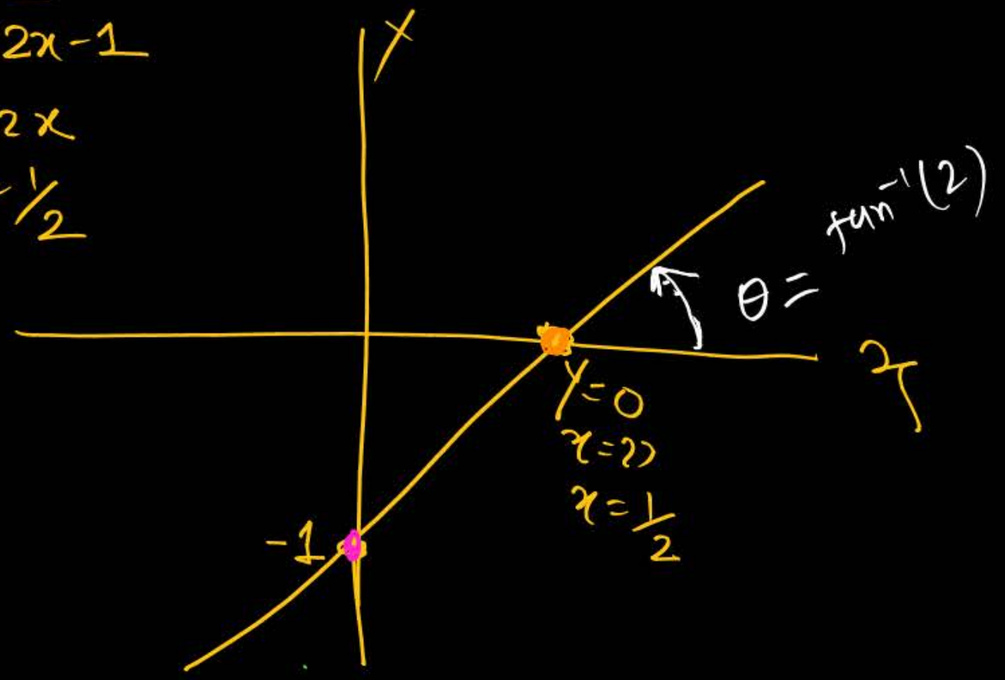
$m = 2 = \tan\theta$

at $y = 0$

$0 = 2x - 1$

$1 = 2x$

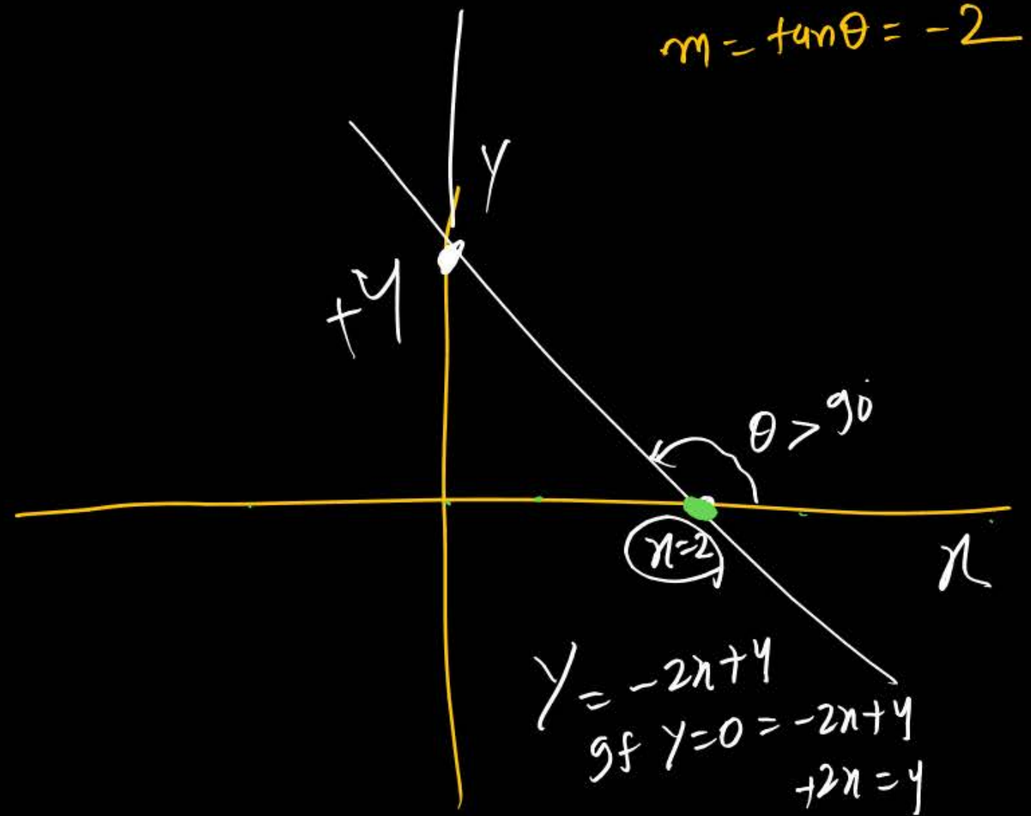
$x = \frac{1}{2}$



$y = -2x + 4$ ← comp. $y = mx + c$

$c = 4$

$m = \tan\theta = -2$



$y = -2x + 4$

if $y = 0 = -2x + 4$

$+2x = 4$

$x = 2$

$$\frac{x}{3} + \frac{y}{4} + 1 = 0$$

$$\frac{y}{4} = -\frac{x}{3} - 1$$

$$y = -\frac{4}{3}x - 4$$

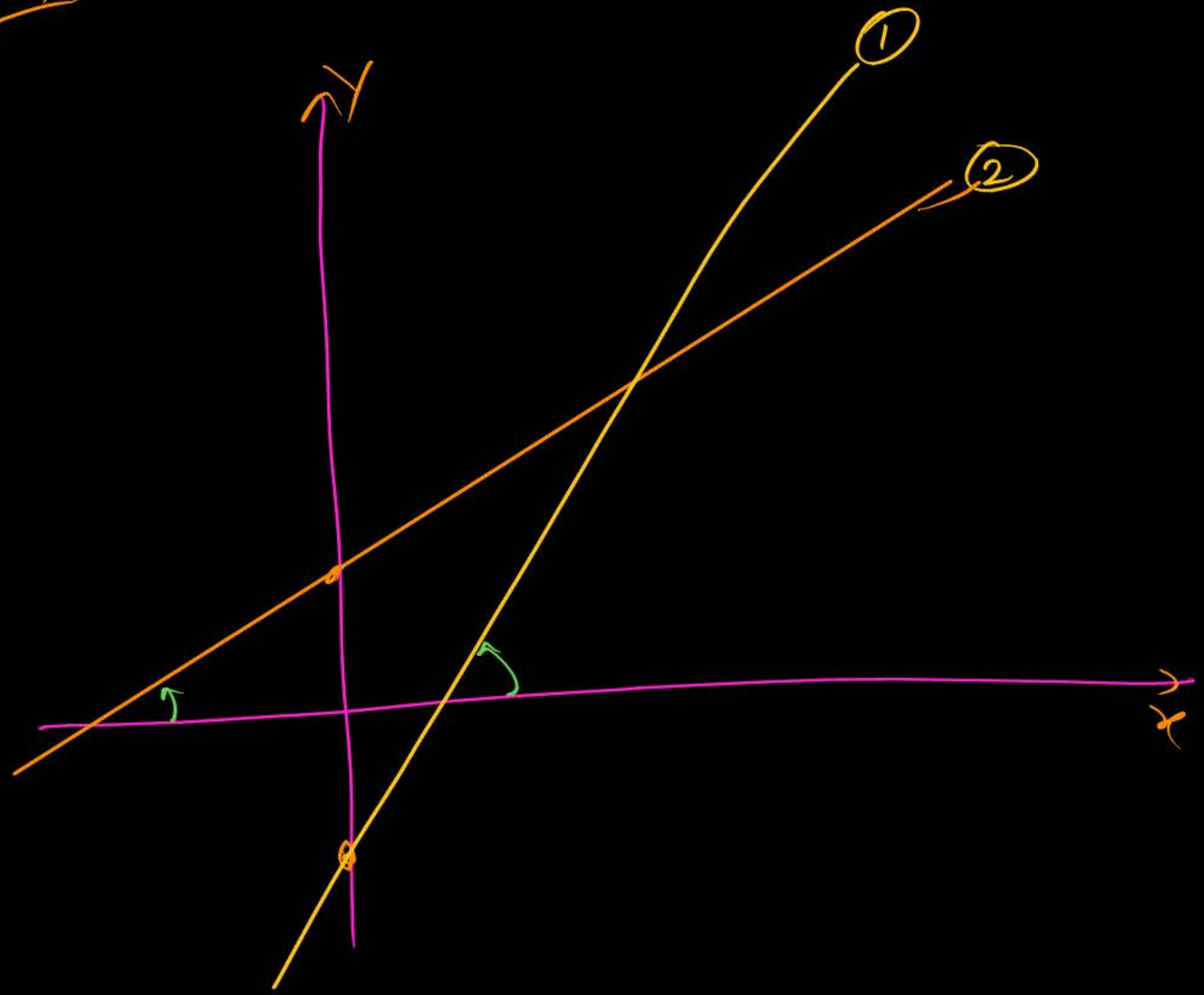
→ compa $y = mx + c$

$$(y-3) + (x-4) + 5 = 0$$

$$y + x = +7 - 5 = 2$$

$$y = -x + 2$$

No copy



Both have +ve slope → True

y intercep of ① +ve → False

" " " ② -ve → False

y intr of ① -ve → True

y interc of ② +ve → True

Question

Draw graph having Y-intercept 4 and passing through (2, 6).

(x, y)

$c = +4$

$y = mx + c$

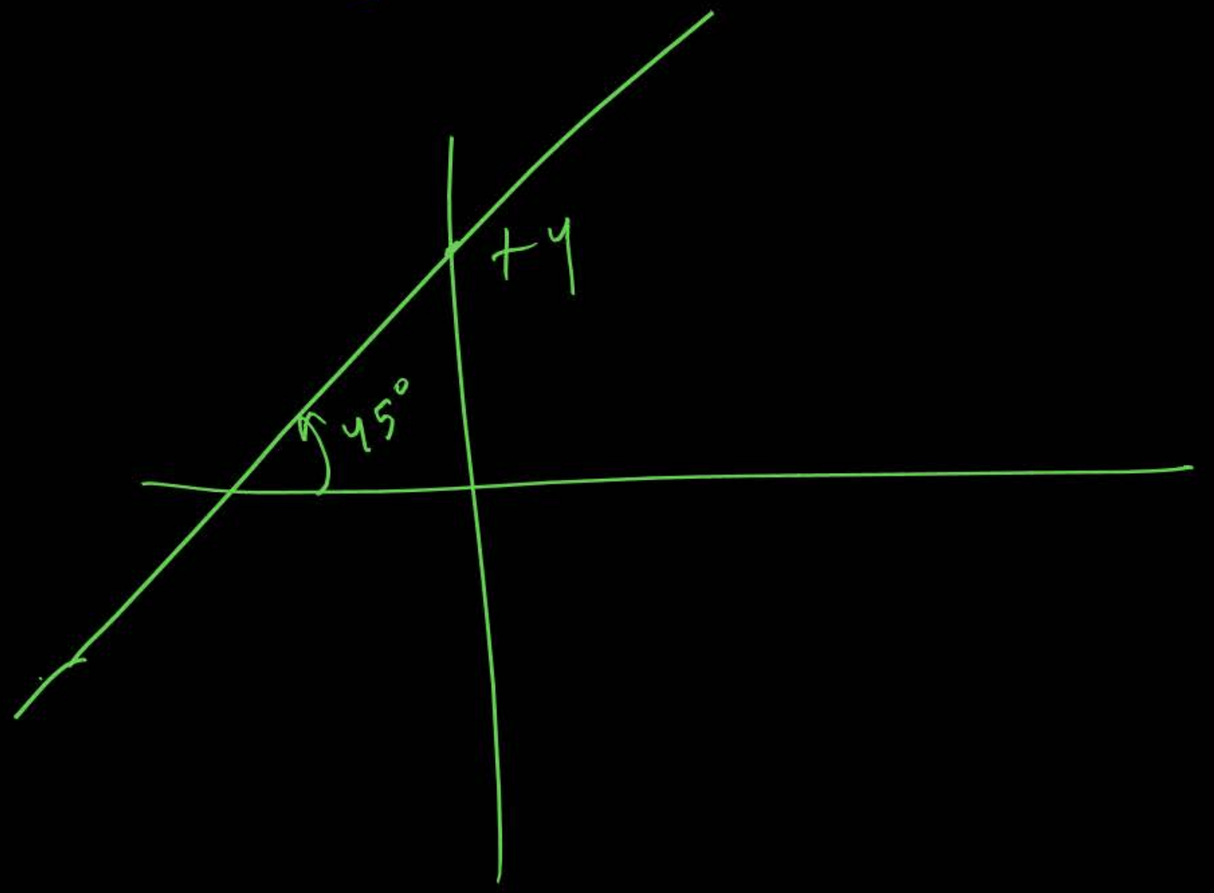
Put $(x, y) = (2, 6)$ give.

$6 = m(2) + 4$

$6 = 2m + 4$

$6 - 4 = 2m$

$m = \frac{2}{2} = 1$. $\checkmark = \tan\theta$



Question

Draw graph passing through (2, 3) and slope 1.

(x, y) $\quad m = 1$

Question

Find equation of a straight line passing through point (3, 4) and (2, 6).

(x_1, y_1) (x_2, y_2)

$\tan\theta = \frac{y_2 - y_1}{x_2 - x_1}$

$\tan\theta = m = \frac{6-4}{2-3}$

$= \frac{2}{-1}$

$\boxed{m = -2}$

$y = mx + c$

$4 = 3m + c \quad - (1)$

$6 = 2m + c \quad - (2)$

$(2) - (1)$

$2m + c = 6$

$3m + c = 4$

$-m = 2$

$\boxed{m = -2}$

Put m in eq^n (1)

$3m + c = 4$

$3(-2) + c = 4$

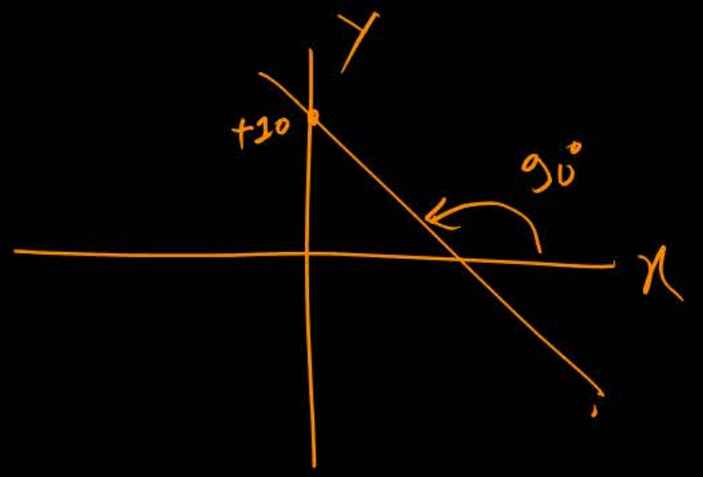
$-6 + c = 4$

$c = 10$

$c = 10$

$m = -2 = \tan\theta$

$\theta > 90$

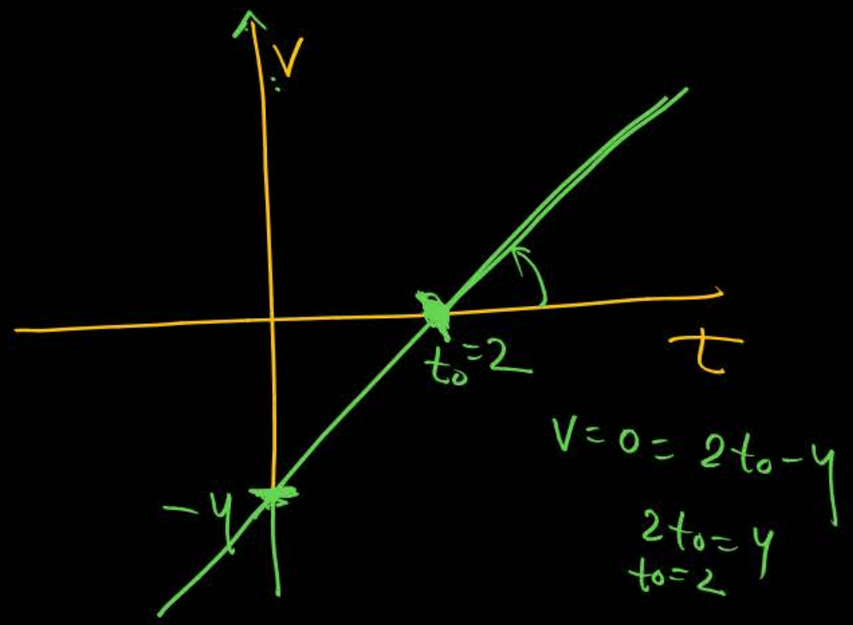


Velocity of object $V = 2t - 4$ then Draw Velocity time graph

$y = mx + c$

$c = -4$

$m = 2 = \tan\theta$



$V = 0 = 2t_0 - 4$

$2t_0 = 4$

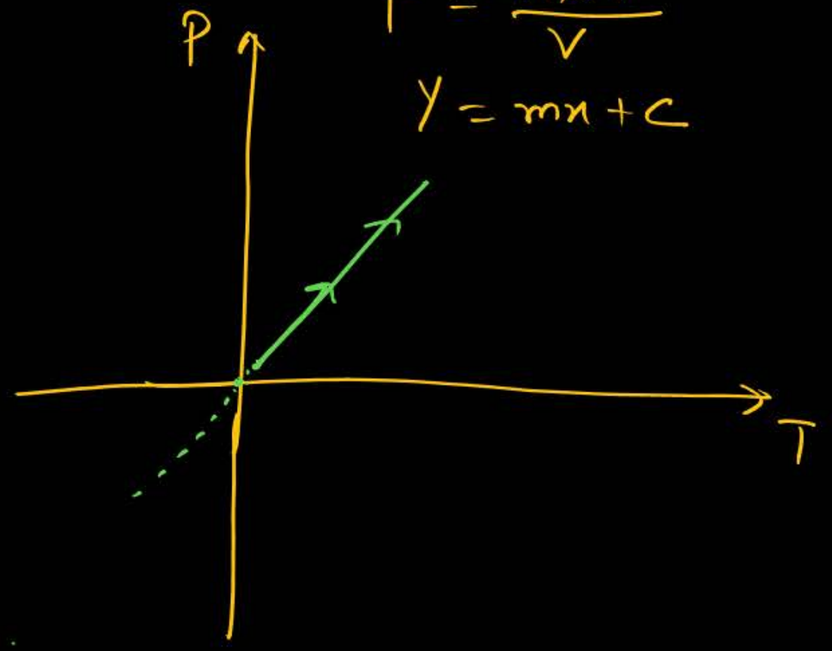
$t_0 = 2$

If $PV = nRT$ then draw "P and T" if V=constant

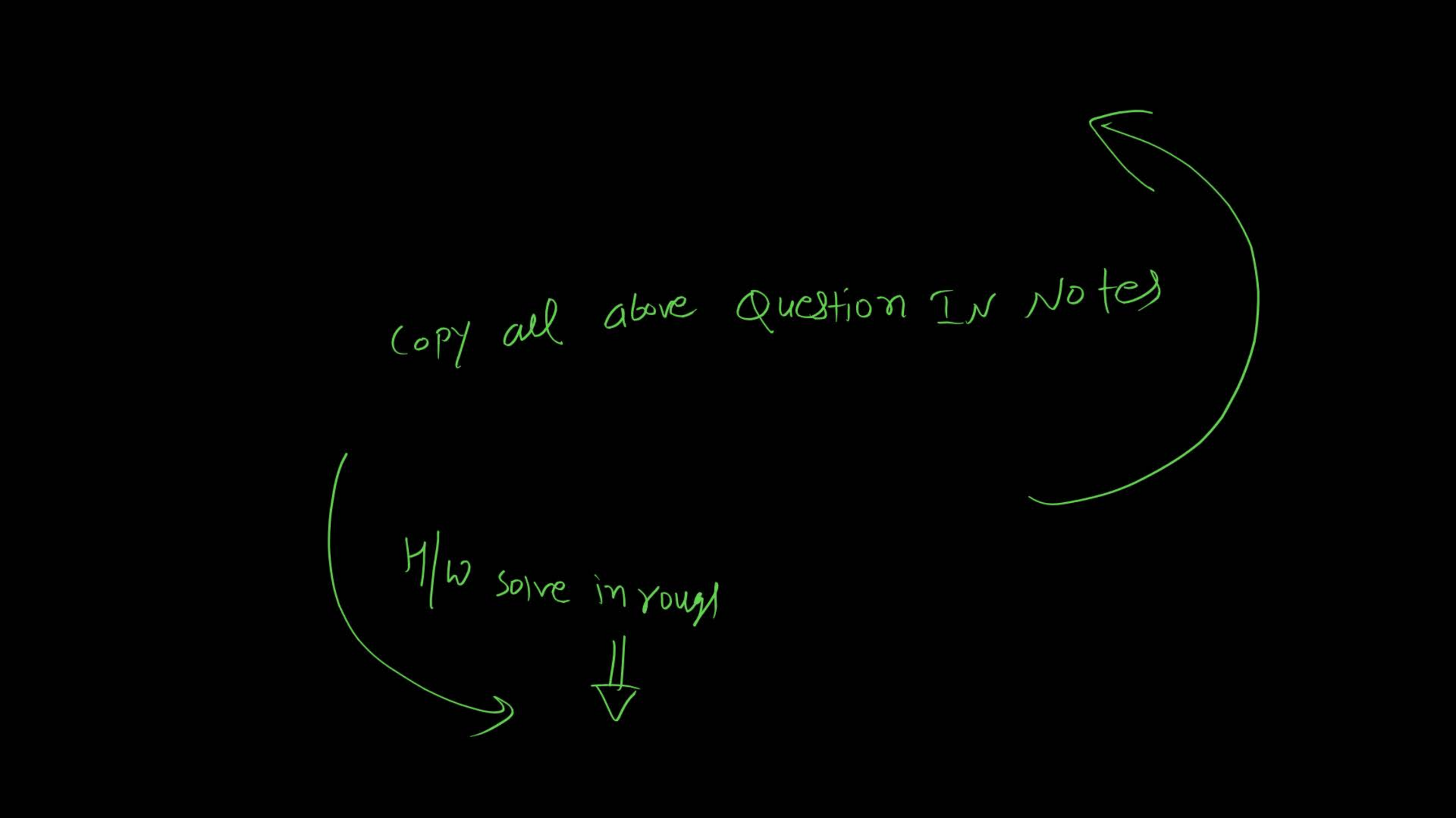
$(P)V = nR(T)$

$P = \frac{nRT}{V}$

$y = mx + c$



If equation $\frac{C-0}{100-0} = \frac{K-273}{373-273}$ then draw graph b/w C & K


copy all above Question IN Notes
H/W solve in rough

Question

Find equation of the line which makes intercept +4 and 5 on the x and y-axis.

1. $5x + 4y + 20 = 0$
2. $4y + 5y - 20 = 0$
3. $4y - 5x = -20$
4. $4x + 5y + 20 = 0$

Question

Draw graph between y and x for given equation:

(i) $y = \sqrt{3}x + 4$

(ii) $y = -x$

(iii) $y = |x|$

(iv) $\frac{x}{3} + \frac{y}{4} + 1 = 0$



Sharsh Assignment 1 & 2

Join

Thank

You